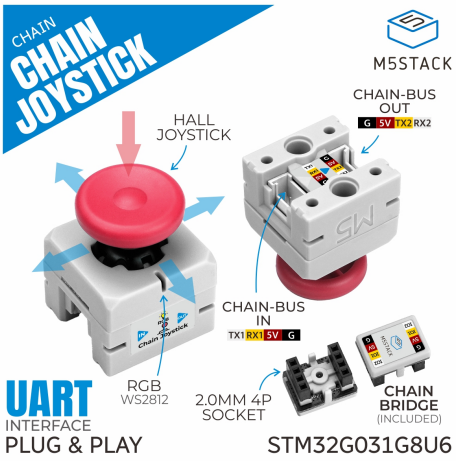
Chain Joystick

SKU:U205























Description

Chain Joystick is a high-precision Hall effect joystick input node in the M5Stack Chain series. It supports three-axis control signal input, including analog input for X/Y axes and digital key input for the Z-axis. The device uses a Hall effect joystick that achieves high-precision control by detecting magnetic field changes. It features non-contact operation, wear resistance, high accuracy, and strong anti-interference capability, ensuring product stability and long lifespan. It also integrates a programmable RGB LED for status indication and interactive display. It is suitable for human-machine interaction, robot control, and other application scenarios.

Chain Joystick integrates the STM32G031G8U6 core controller and adopts the UART serial cascading communication protocol. Through two HY2.0-4P expansion interfaces, it can extend more devices from the Chain series to build richer interactive applications.

Features

- M5Stack Chain series
- High-precision Hall effect joystick
- STM32G031G8U6 core controller
- X/Y/Z three-axis input
- 1x RGB LED
- UART serial cascading communication protocol
- 2x HY2.0-4P expansion interfaces supporting Chain series extensions

Includes

- 1 x Chain Joystick
- 1 x Chain Bridge

Applications

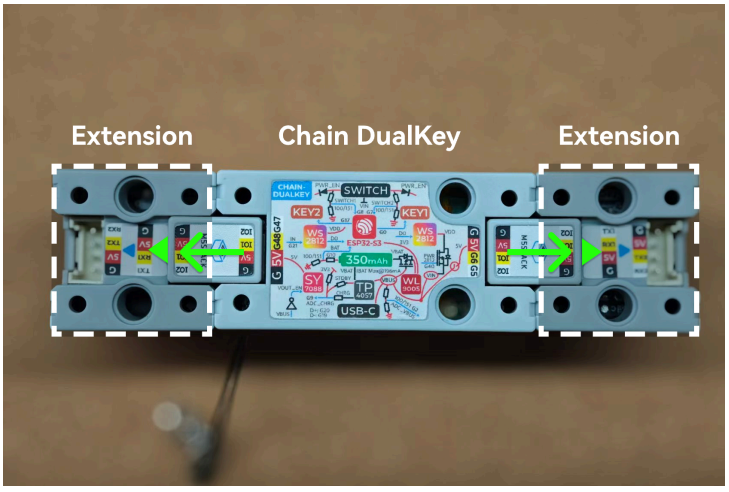
- Robot control
- Human-machine interaction

Specifications

Specification	Parameter
MCU	STM32G031G8U6
Input Power	DC 5V
Communication Method	UART 115200bps@8N1
Interface Type	2x HY2.0-4P
RGB LED	1x WS2812C
Operating Power Consumption	25.4mA
Operating Temperature	0 ~ 40°C
Product Size	23.9 x 23.9 x 30.4mm
Product Weight	8.6g
Package Size	138.0 x 93.0 x 25.0mm
Gross Weight	12.0g

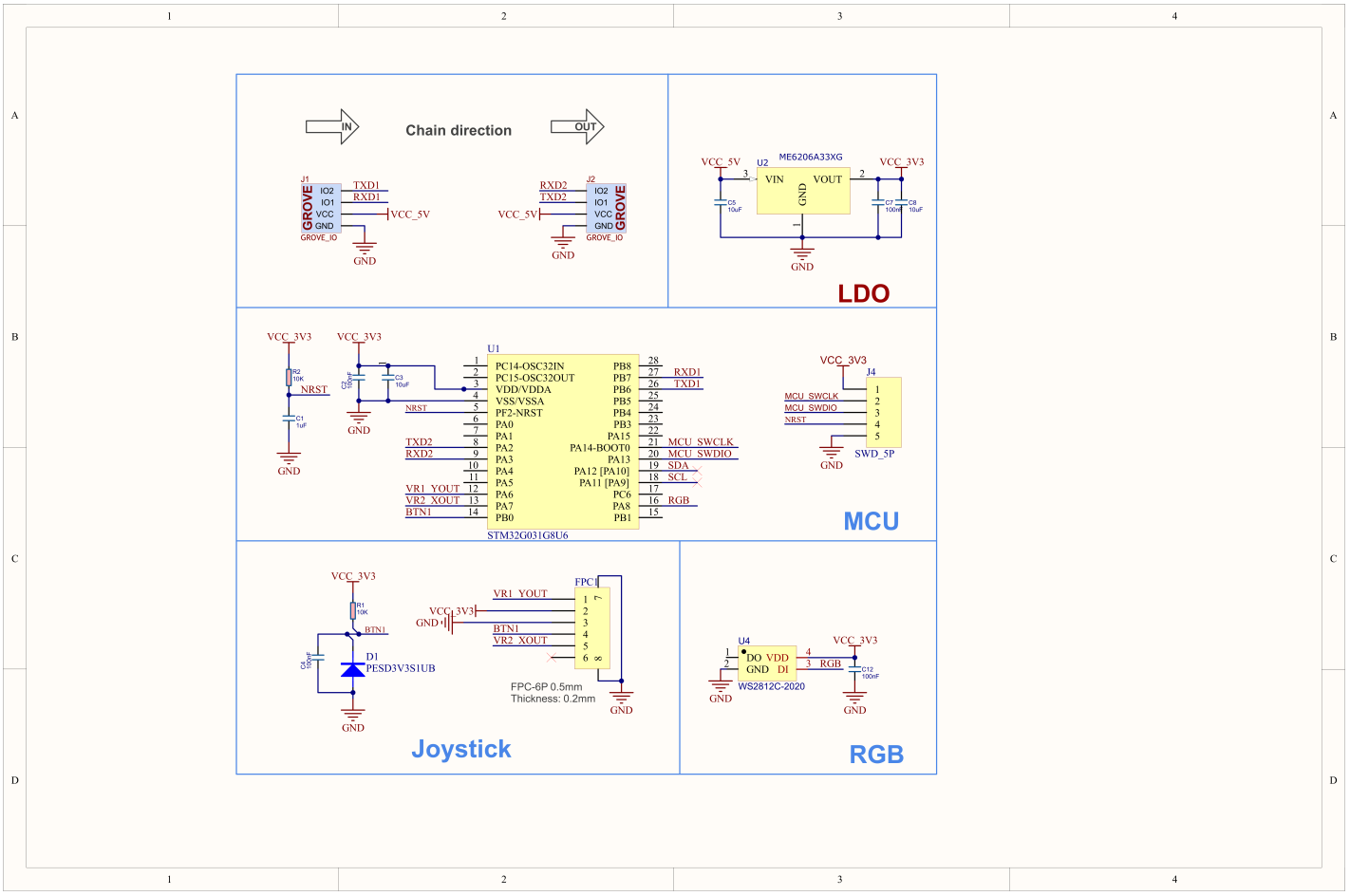
Learn

Use the Chain Bridge connector to connect the main controller Chain DualKey and various Chain series input devices. Pay attention to the direction when connecting — the triangle arrow should point outward from the main controller Chain DualKey, as shown below:



Schematics

- Chain Joystick Schematics PDF



PinMap

RGB LED

STM32G031	PA8
WS2812C	RGB

Joystick

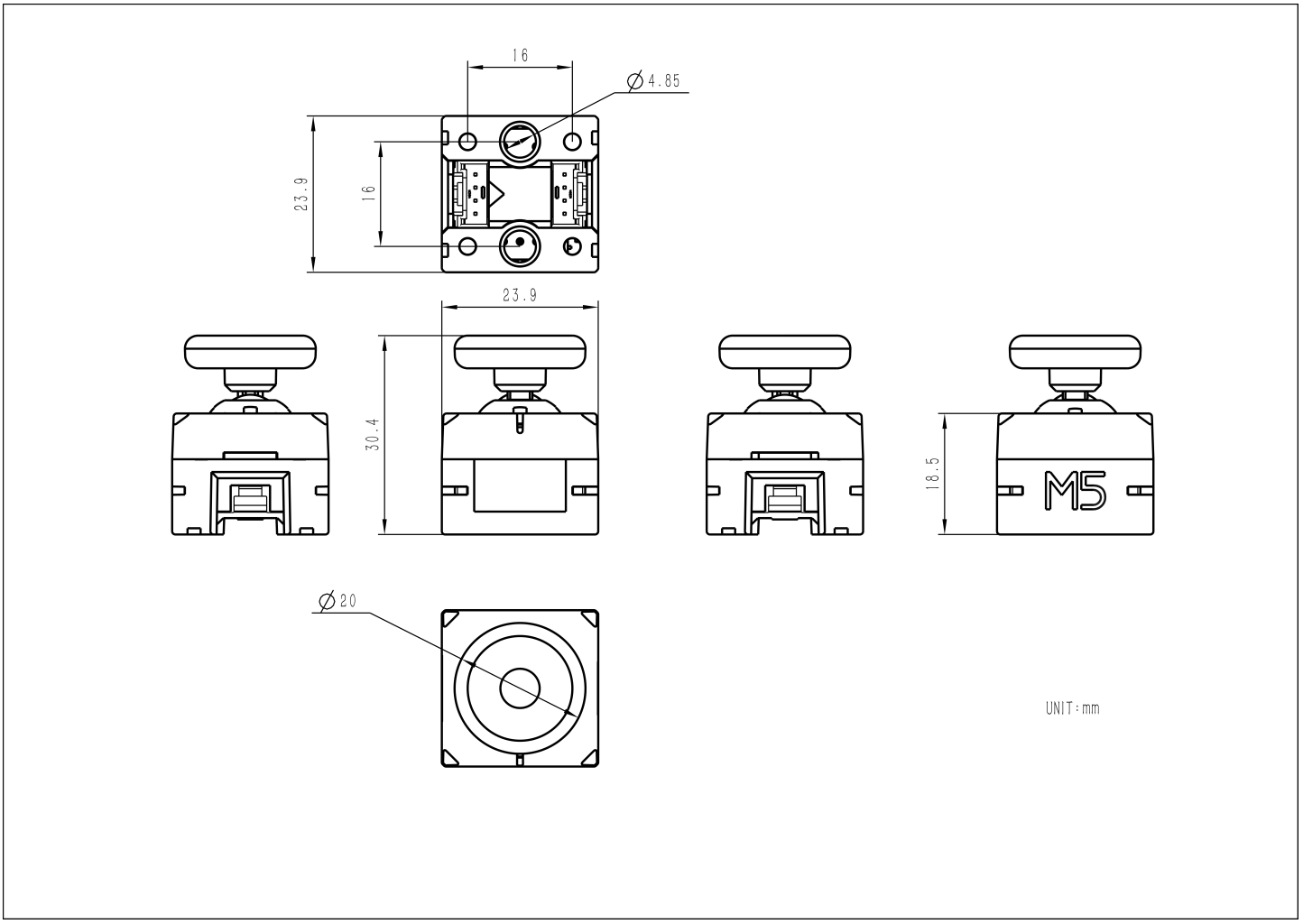
STM32G031	PB0	PA7	PA6
Button	Input	XOUT	YOUT

UART

STM32G031	PB6	PB7	PA2	PA3
UART1	TXD1	RXD1		
UART2			TXD2	RXD2

Model Size

- Chain Joystick Model Size PDF



Softwares

Arduino

- Chain Joystick Arduino Quick Start
- Chain Joystick Arduino Library
- Chain Joystick Example Program

UiFlow2

- Chain Joystick UiFlow2 Docs

Internal Firmware

- Chain Joystick Internal Firmware

Protocol

View Protocol

Video

- Chain Joystick Product Introduction and Function Demonstration

U205-Chain-Joystick-video-EN.mp4